STRATEGIC INSIGHTS REPORT

Credit Risk & Customer Financial Health Analytics

An evidence-led monitoring and customer-support framework

Prepared for the Executive Board

FinTrust Digital Finance

ITS 2122 - Python for Data Science & AI | Semester 3, 2026

IJSE - Institute of Software Engineering

Group and Student Information

Replace all bracketed placeholders before submission; remove the fifth row if the group has four members.

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AI-Assistance Declaration

AI-assisted tools were used to support document structuring, language refinement, code review, chart preparation, and generation of the editable Word report. All data-cleaning decisions, calculations, analytical findings, visualisations, interpretations, references, and recommendations were reviewed and validated by the project team before submission.

The project team accepts responsibility for the accuracy, originality, and final content of this submission.

Executive Summary

FinTrust Digital Finance needs a consistent way to detect repayment stress, direct collections capacity, and support customers before arrears become severe. This report analyses 30,000 historical credit-card accounts from Taiwan. 6,636 customers defaulted in the following month, producing a portfolio default rate of 22.12%. The evidence supports behaviour-led monitoring: recent repayment delay and weak payment coverage are more useful operational signals than demographic characteristics.

Repeated delay is the clearest warning pattern. Customers with no delayed months recorded an 11.7% default rate, compared with 70.3% among customers delayed in all six months. Financial stress is especially concentrated when utilisation is at least 50% and the payment-to-bill ratio is below 20%. That group contains 10,378 customers (34.6% of the portfolio) and 3,021 defaults (45.5% of all defaults), with a 29.1% within-group default rate.

The four risk segments provide an interpretable operating language. Critical is the most severe segment, with 1,867 customers and a 59.5% observed default rate. However, At-Risk is the main financial-priority pool: 8,238 customers carry NTD 619.7 million of outstanding exposure and NTD 198.2 million of simplified expected loss, or 53.4% of the portfolio estimate. Severity and absolute exposure therefore require different resource decisions.

Three actions should be prioritised. Customer Engagement and CRM should send Watchlist reminders three to five days before the due date and refresh segments monthly. Credit Risk and Collections should contact At-Risk customers within five business days, assess current behaviour and affordability, and offer structured repayment support before considering a limit change. Collections and Hardship Support should assign Critical cases within one business day and review them weekly. These are decision-support recommendations, not automatic adverse actions: demographic attributes must remain descriptive, and any restriction requires human review, a transparent reason, and a route for the customer to respond.

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1. Introduction

FinTrust Digital Finance is presented in the project brief as a growing lender whose monitoring and collections decisions remain largely manual. The business requirement is to convert six months of account behaviour into evidence that supports earlier intervention, more disciplined prioritisation, and fairer customer contact. The analysis therefore asks where default risk is concentrated, which repayment patterns offer useful warning, how exposure differs from severity, and how customer groups should be managed without turning descriptive associations into automatic decisions.

This report is written for decision-makers rather than programmers. It summarises the validated technical appendix, distinguishes observation from interpretation, and treats the resulting score as an academic prioritisation framework. It does not estimate a calibrated probability of default and should not be deployed as a production credit engine without further validation, governance, and legal review.

2. Data and Methodology

2.1 Dataset Scope and Data Quality

The source is the UCI Default of Credit Card Clients dataset, originally assembled from Taiwanese credit-card accounts and associated with the study by Yeh and Lien (2009). It contains 30,000 customers, 23 explanatory variables, an identifier, and a next-month default target. The records cover credit limits, demographic codes, six monthly repayment-status fields, six bills, and six payments. All monetary values are New Taiwan Dollars (TWD/NTD), and the observation period is historical rather than current.

The raw file required the second row as the true header and was validated at 30,000 rows by 25 columns. The cleaning audit found no raw nulls, duplicate rows, duplicate identifiers, impossible ages, non-positive credit limits, or negative payment amounts. Sex, education, and marital-status codes were decoded into readable labels; undocumented codes were retained as Unknown rather than removed. Negative bills were retained because they may represent customer credit balances. The final cleaned file has 30,000 rows and 34 columns.

Three flags preserve unusual but potentially informative records: 1,312 delay-with-no-bill accounts, 3,931 accounts exceeding their limit in at least one month, and 1,930 credit-balance accounts. Six behavioural features were engineered: average bill, average payment, payment-to-bill ratio, delayed-month count, maximum delay, and six-month balance trend. The payment-to-bill ratio is intentionally null for 1,071 customers whose average bill is zero or negative; a denominator in that state does not describe repayment capacity.

2.2 Analytical Framework and Assumptions

Exploratory analysis used grouped rates, distributions, and a focused Pearson correlation matrix. Age bands use fixed ranges, while credit-limit tiers use portfolio quartiles. These tiers are descriptive comparisons, not permanent policy thresholds. Continuous scoring dimensions use equal-frequency quintiles implemented with pandas qcut (pandas development team, 2026), so they rank customers relative to this portfolio and must be recalibrated if its distribution changes.

The score combines four dimensions. Delinquency uses fixed rules because two-thirds of customers have no delayed months; it receives double weight because delay frequency has the strongest observed relationship with default. Repayment capacity reverses the payment-ratio quintiles so lower coverage means higher risk; the 1,071 undefined ratios receive neutral score 3 only for this dimension. Exposure uses the average bill divided by credit limit, clipped at zero, and trajectory measures whether the recent bill is larger than the oldest bill. Scores range from 5 to 25 and map to Healthy (5-9), Watchlist (10-14), At-Risk (15-19), and Critical (20-25).

Table 1. Interpretable risk-scoring framework used for academic segmentation.

Dimension	Measure	Scoring direction and method	Weight
Delinquency	Delayed months	Fixed rules; repeated delay scores higher	2
Repayment capacity	Payment-to-bill ratio	Reversed quintiles; lower coverage scores higher	1
Exposure	Average bill / credit limit	Quintiles; higher utilisation scores higher	1
Trajectory	Recent bill - oldest bill	Quintiles; faster balance growth scores higher	1

Validation requires observed default rates to increase across the four segments. A separate override places 495 strictly defined ghost or dormant accounts in At-Risk because their one recent delay and five inactive months produce misleadingly low exposure scores despite a 39.6% observed default rate. Phase 4 defines outstanding exposure as positive average bill and estimates expected loss as segment exposure multiplied by observed segment default rate. This simplified estimate assumes 100% loss given default and no recovery; it is for prioritisation, not accounting provision.

3. Credit Risk and Repayment Analysis

3.1 Portfolio and Descriptive Risk Profile

The portfolio contains 23,364 non-defaulting and 6,636 defaulting accounts, giving a 22.12% baseline. Figure 1 establishes the denominator for later comparisons. A rate above the baseline identifies higher observed risk, but the number of customers and the value exposed also matter when deciding where to allocate effort.

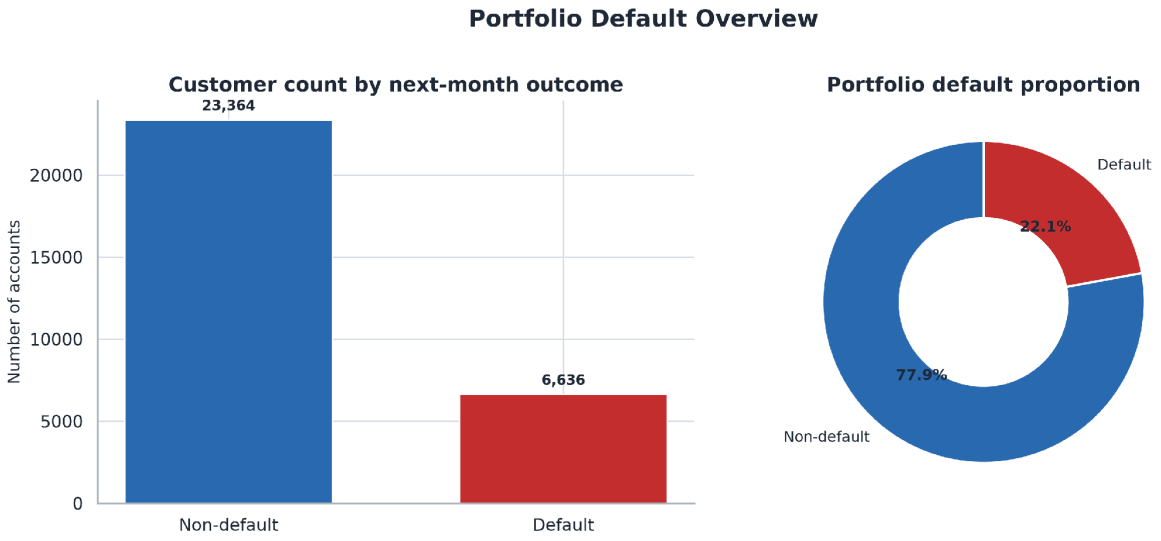


Figure 1. Portfolio default overview.

Credit-limit quartiles show the strongest descriptive group contrast. The Low tier contains 7,676 customers and defaults at 31.8%, while the Very High tier contains 7,067 customers and defaults at 14.0%. Figure 2 does not show that a lower limit causes default: limits may reflect prior underwriting, customer circumstances, or other omitted factors. The result identifies a group for behavioural review, not a rule for adverse treatment. Similarly, the 60+ age band has a 28.3% rate but only 339 customers, so it is descriptive and comparatively uncertain.

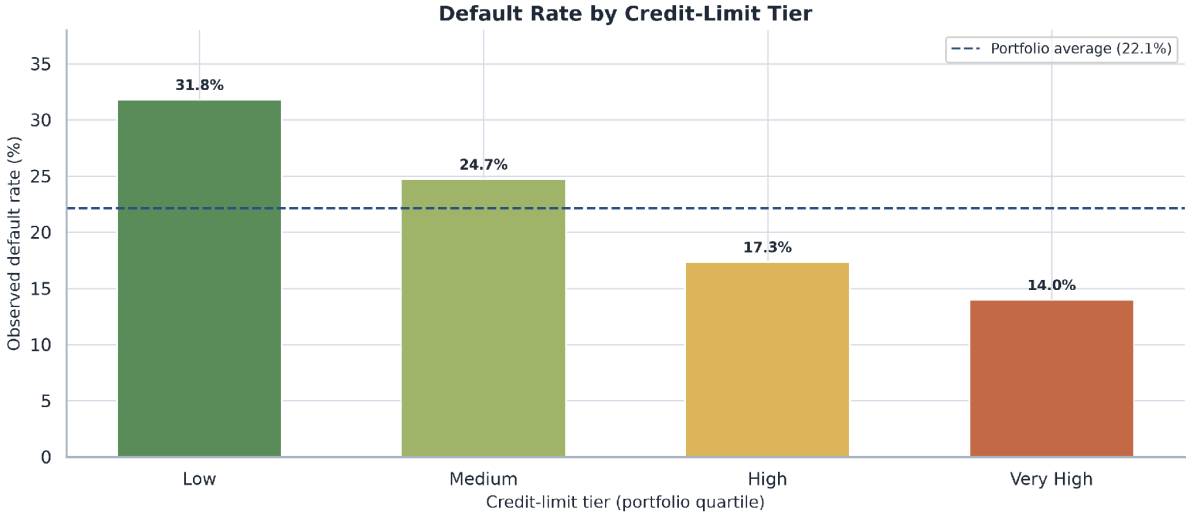


Figure 2. Default rate by credit-limit tier.

3.2 Repayment Delay as the Primary Warning Signal

Delay frequency provides the clearest progression. Customers with no delayed months default at 11.7%; one delayed month raises the rate to 29.8%, three delayed months to 50.9%, and six delayed months to 70.3%. The rise is near-monotonic, with a plateau between four and five months before the six-month

group increases again. This evidence supports monthly monitoring and the double weight assigned to delinquency, while avoiding a claim that delay alone explains every default.

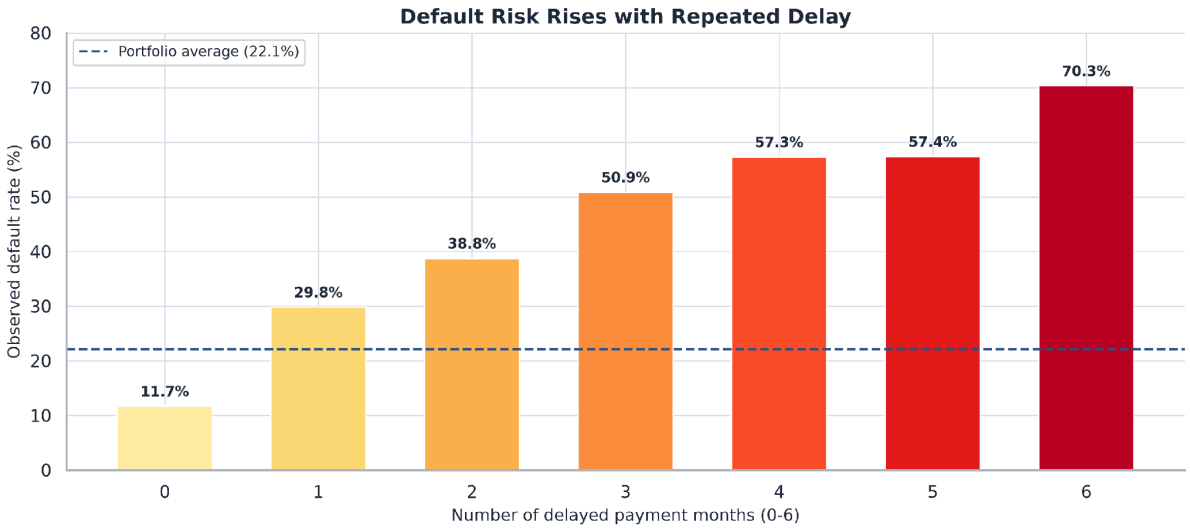


Figure 3. Default rate by delayed payment months.

3.3 Payment Capacity and Financial Behaviour

Figure 4 corrects a common but unsupported reading of the bills. Defaulted customers have a slightly lower median average bill (NTD 19,781) than non-defaulted customers (NTD 21,443); bill size alone is therefore not the main story. The stronger separation is repayment: median average payment is NTD 1,612 for defaulted customers versus NTD 2,754 for non-defaulted customers, and the median payment-to-bill ratio is 0.06 versus 0.12. Outliers are omitted only from the visual, not from the analysis.

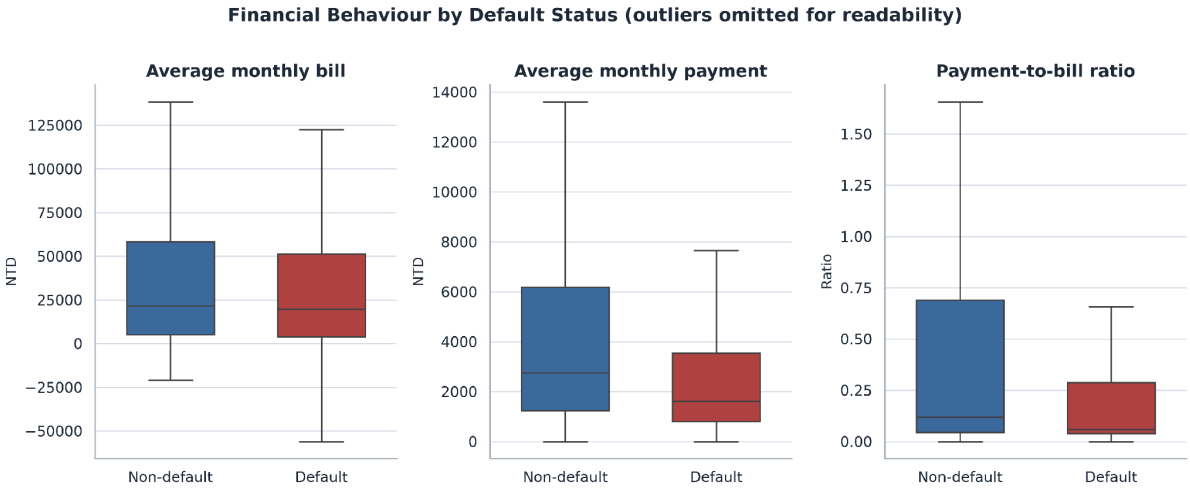


Figure 4. Financial behaviour by default status.

The focused correlation output reinforces that interpretation. Delayed-month count has a 0.40 Pearson correlation with default, maximum delay 0.37, and the most recent payment-status field 0.33. Average bill and payment-to-bill ratio each have correlations close to zero with the binary target when considered

alone. These are associations, not causal effects, but they justify using repeated behaviour rather than a single balance figure as the first monitoring signal.

A material blind spot remains. The No-Warning-Signal archetype includes 17,326 customers with no recorded delays and utilisation below 80%; 1,951 still defaulted. That is 29.4% of all defaults, although the archetype's own rate is only 11.3%. The observed six-month behaviour cannot distinguish those cases reliably. FinTrust should manage this as a portfolio-level uncertainty and seek better affordability evidence, rather than manufacturing individual warnings from weak demographic proxies.

4. Customer Risk Segmentation

4.1 Segment Validation and Interpretation

The segment results are monotonic: Healthy defaults at 13.9%, Watchlist at 15.0%, At-Risk at 32.0%, and Critical at 59.5%. Figure 5 shows both size and severity. The narrow Healthy-Watchlist gap is intentional: both groups are dominated by customers without delay, while Watchlist captures weaker secondary conditions such as lower payment coverage, higher utilisation, or an adverse balance trajectory. Its value is preventive outreach, not a claim of severe current delinquency.

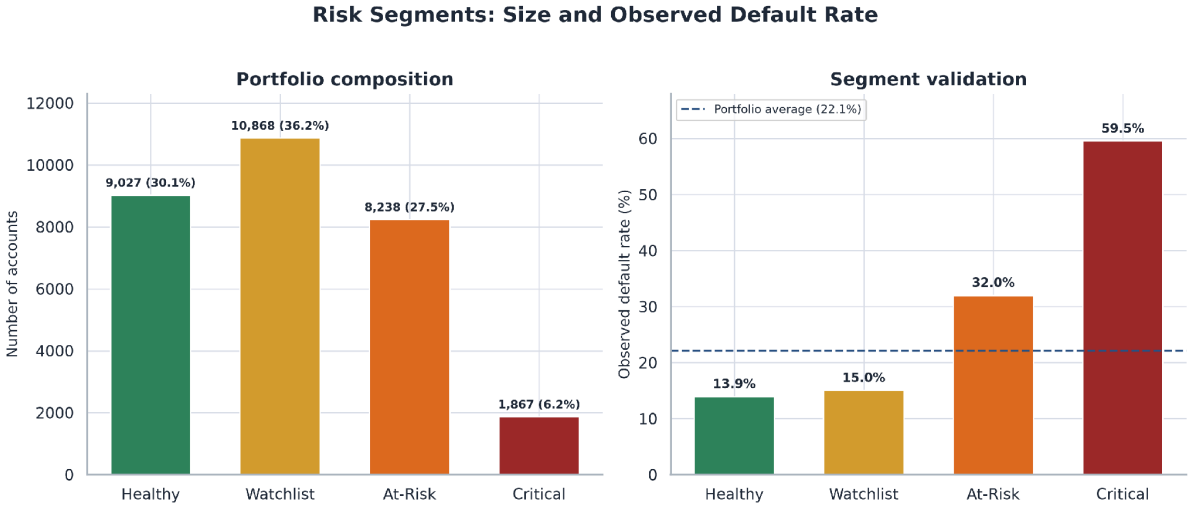


Figure 5. Risk-segment composition and validation.

Table 2. Segment validation, exposure, and simplified expected-loss summary.

Segment	Accounts	Share	Default rate	Exposure	Expected loss
Healthy	9,027	30.1%	13.9%	NTD 108.4M	NTD 15.1M (4.1%)
Watchlist	10,868	36.2%	15.0%	NTD 477.4M	NTD 71.7M (19.3%)
At-Risk	8,238	27.5%	32.0%	NTD 619.7M	NTD 198.2M (53.4%)
Critical	1,867	6.2%	59.5%	NTD 144.2M	NTD 85.8M (23.1%)

Behavioral profiles explain the labels. Healthy averages 0.1 delayed months, a 0.91 median payment ratio, and 0.06 utilisation. Watchlist averages 0.3 delayed months, 0.09 median payment ratio, and 0.39 utilisation. At-Risk rises to 1.4 delayed months and 0.60 utilisation; Critical reaches 4.5 delayed months

and 0.79 utilisation. The score is therefore a structured summary of observable behaviour, not a substitute for an individual case assessment.

4.2 Ghost and Dormant Account Overlay

The strict ghost definition identifies 495 accounts with five inactive months, one recent delay flag, no bills, and no payments. 196 defaulted (39.6%), yet the group has NTD 0 outstanding exposure under the project definition. The override keeps these accounts visible in At-Risk, but the operating response must be account-status confirmation or re-engagement. It would be inappropriate to infer affordability stress or impose a standard restructuring path when no balance is owed.

5. Credit Exposure and Strategic Recommendations

5.1 Utilisation and Repayment Interaction

Figure 6 separates utilisation from repayment behaviour using fixed, business-readable thresholds. The high-utilisation and poor-repayment quadrant contains 10,378 customers and defaults at 29.1%. It contributes 3,021 defaults, or 45.5% of all defaults, despite representing 34.6% of the portfolio. By contrast, the 681 high-utilisation customers with adequate repayment default at 18.4%. High utilisation should therefore prompt context-sensitive review, not automatic restriction. The 1,071 undefined payment-ratio records are excluded from this two-variable test and handled separately.

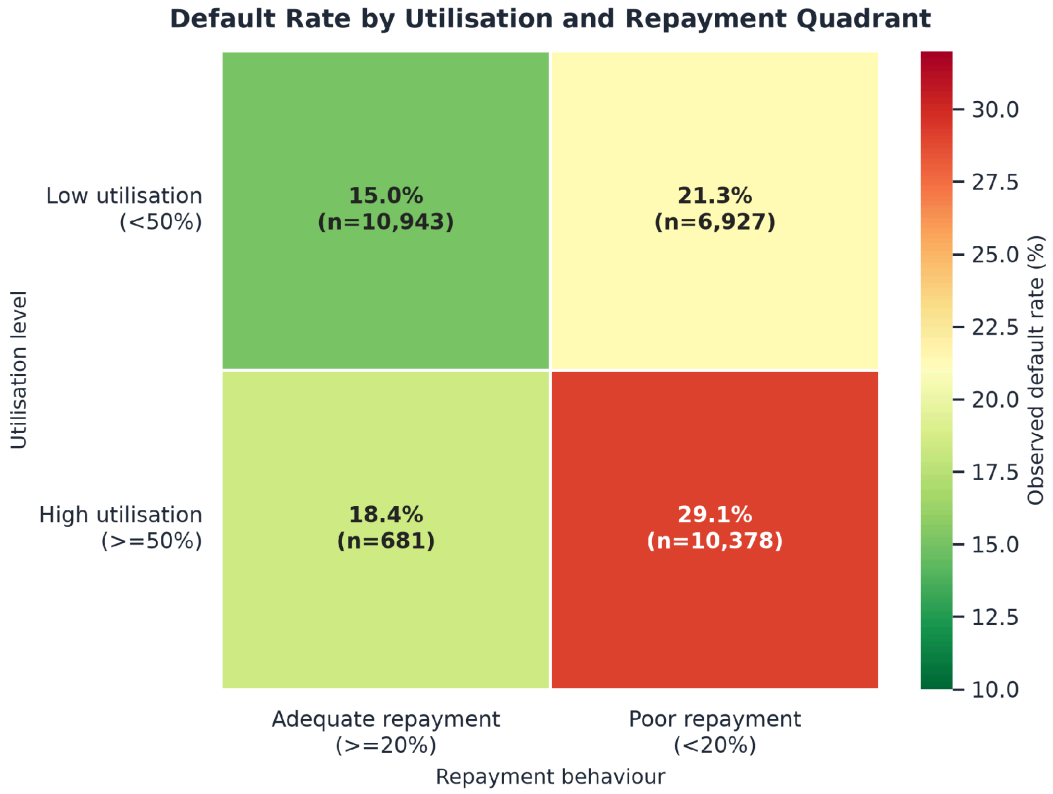


Figure 6. Utilisation and repayment quadrant.

5.2 Exposure and Expected-Loss Priority

Critical has the highest rate, but At-Risk carries the largest money pool. At-Risk holds NTD 619.7 million of outstanding exposure and NTD 198.2 million of simplified expected loss (53.4% of the total). Critical holds NTD 144.2 million and NTD 85.8 million (23.1%). Figure 7 supports a dual-capacity model: scale preventive work for At-Risk while reserving intensive case management for the smaller Critical segment.

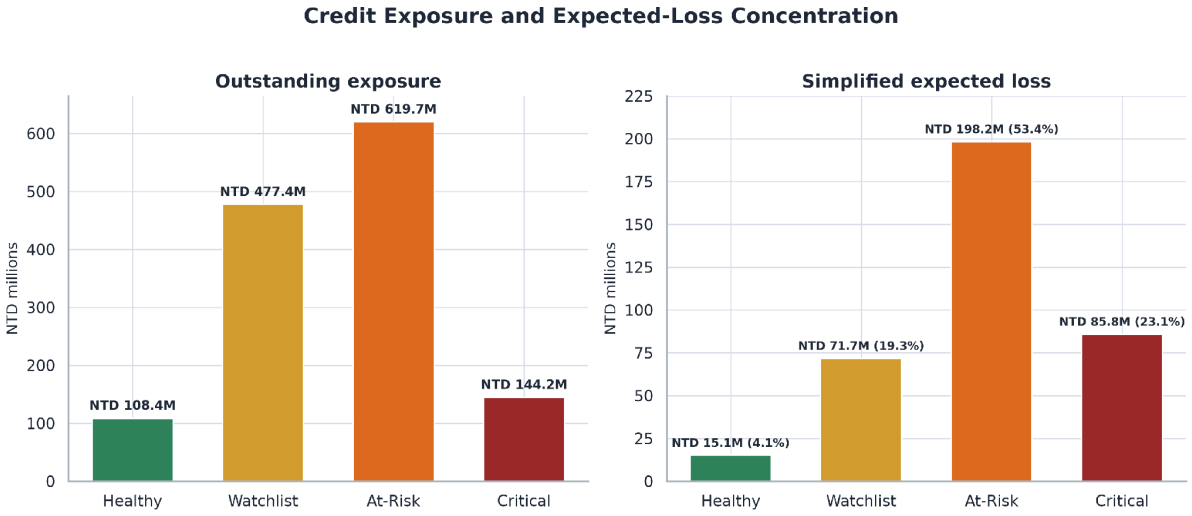


Figure 7. Exposure and expected loss by segment.

5.3 Priority 1: Watchlist Early-Warning Journey

Target Watchlist customers at the monthly segment refresh. Customer Engagement and CRM should send a clear reminder three to five days before the due date, provide optional financial-wellness support, and reassess the segment monthly. Success measures should include on-time payment, movement to Healthy, and the incidence of a first delayed month. The safeguard is equally important: the message is supportive, and Watchlist status alone must not trigger a limit reduction or collections escalation.

5.4 Priority 2: At-Risk Exposure Review

Credit Risk and Collections should create an At-Risk queue after each monthly refresh and contact customers within five business days. Staff should review current exposure, recent repayment, affordability, and hardship circumstances, then offer a structured repayment plan or another proportionate adjustment where appropriate. A credit-limit change requires individual evidence and a recorded reason. Management should track contact completion, plan uptake, segment migration, and changes in the expected-loss pool.

5.5 Priority 3: Critical Case Management

Collections and Hardship Support should assign each Critical case within one business day and review it weekly. The objective is sustainable resolution: verify affordability, consider hardship restructuring or settlement, and restrict further exposure only after human review. Track time to contact, affordability-assessment completion, plan adherence, cure, complaints, and repeat escalation. For

Healthy customers, maintain standard service; for ghost accounts, use account-status communication rather than conventional collections.

6. External Data Enrichment

Phase 5 calls the public ExchangeRate-API Open Access endpoint with TWD as the base currency. The saved final-notebook response reports 1 TWD = 0.031076 USD at Tue, 14 Jul 2026 00:02:31 +0000 (ExchangeRate-API, 2026). The request validates HTTP status, result, base code, rate, and timestamp, and handles timeouts, connection failures, status errors, and invalid payloads. If validation fails, no USD values are created; no private key or fabricated fallback is used.

Table 3. Validated Phase 5 reporting snapshot from the executed technical appendix.

Metric	TWD value	USD reporting value
Exchange rate	1 TWD	USD 0.031076
Total credit limit	NTD 5.025B	USD 156.1M
Outstanding exposure	NTD 1.350B	USD 41.9M

At that dated rate, NTD 5.025 billion of total credit limit is approximately USD 156.1 million, while NTD 1.350 billion of outstanding exposure is approximately USD 41.9 million. The enrichment helps international stakeholders read portfolio scale in a familiar currency. It is a current reporting snapshot, not a historical conversion, pricing signal, or customer-risk input.

7. Ethics, Fairness, and Limitations

7.1 Responsible Use

Age, sex, education, and marital status are used only to describe portfolio patterns and monitor outcomes. They must not trigger automatic limit reductions, collections escalation, or different services. Any action affecting credit access should be based on documented behaviour and current circumstances, reviewed by a responsible employee, explained in understandable language, and accompanied by a contact or challenge route. Segment performance and treatment outcomes should be monitored for drift and unfair impact, with decisions retained for audit.

Automated segmentation can improve consistency, but it can also turn historical patterns into self-reinforcing treatment if used without review. The recommended workflow therefore treats segment labels as prompts for support and investigation. Watchlist outreach remains non-adverse, At-Risk changes require individual review, and Critical handling includes affordability and hardship assessment. No demographic variable substitutes for missing evidence about a customer's capacity or reason for distress.

7.2 Analytical and External-Data Limitations

The dataset omits income, employment, household obligations, affordability, reasons for non-payment, recoveries, and post-default outcomes. It covers a historical Taiwanese portfolio, so rates and thresholds may not transfer to FinTrust's current customers or another jurisdiction. Correlation does not establish

causation, and the score is not a calibrated probability-of-default model. Portfolio-relative quintiles can move as the customer mix changes. The simplified expected-loss calculation assumes 100% loss given default and no recovery, so it must not be presented as an accounting estimate.

The exchange-rate result is one daily API snapshot, not a rate matched to the historical account months. Availability, rate limits, and returned values can change. The high number of No-Warning-Signal defaults also shows that six months of account behaviour is incomplete. Future work should validate the framework in a later time period, add carefully governed affordability and recovery data, test fairness by demographic group without using those attributes as adverse triggers, and compare the academic score with a properly calibrated model under formal governance.

8. Conclusion

The analysis supports a graduated, behaviour-led operating model. Repeated delay is the strongest warning; low payment coverage makes high utilisation more concerning; and At-Risk, rather than Critical, contains the largest expected-loss pool. FinTrust should combine light-touch prevention, scaled exposure review, and intensive hardship-aware case management. The framework adds decision discipline only when its assumptions remain visible, demographic attributes stay descriptive, and human accountability governs any adverse outcome.

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